Steady-State Analysis of the Series Resonant DC–DC Converter in Conjunction with Loosely Coupled Transformer—Above Resonance Operation

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***Abstract*— This paper presents the steady-state analysis of the series resonant dc–dc converter (SRC) in conjunction with loosely coupled transformer (LCT) operating above resonance $(f_s/f_o > 1)$. It is an extension of the conventional steady-state analysis of the SRC. Three operational modes other than the continuous current mode (CCM) have been identified, and the boundaries of all the operational modes are defined. Based on the well-defined operational plane, all the operational modes are analyzed. Results are given in normalized form and experimentally verified.**

***Index Terms*— Electric vehicle (EV), inductive charger, series resonant converter (SRC).**

NOMENCLATURE

A. Symbols

C_r	Resonant capacitance.
d	Symmetry coefficient of the transformer.
E_i	Net amount of energy into the converter during one-half switching cycle.
E_o	Energy into the load during one-half switching cycle.
E_e	Total amount of energy exchange between the input voltage source V_i and the inverter input.
f_o	Load-resonant frequency.
f_o'	Transformer-resonant frequency.
f_s	Normalized switching frequency.
$I_1(t)$	Primary current at time t.
$I_2(t)$	Secondary current at time t.
$I_{\rm rms}$	Primary rms value current.
k	Coupling coefficient of the transformer.
K_q	Term defined by (14).
L_1	Primary inductance of the transformer.
L_2	Secondary side inductance of the transformer.
L_{d1}	Primary leakage inductance.
L_{d2}	Secondary leakage inductance.
L_m	Magnetizing inductance of the transformer.
L_r	Resonant inductance.
L_{s1}	Primary side inductance of the transformer while the secondary is short circuited.
Q	The quality of the load resonant loop.
q	Normalized output voltage or the conversion ratio.
q'	Transformed conversion ratio, defined by (14).
T_s	Time interval of one-half switching cycle.
$V_c(t)$	Voltage on the resonant capacitor at time t.
V_i	Input voltage source.
V_o	Output voltage sink.
Z_o	Characteristic resonant impedance.
α	Time angle for the circuit mode $sc(1,1)$ to appear, measured by ω_o.
β	Time angle for the circuit mode $sc(1,-1)$ to appear, measured by ω_o.
δ	Phase difference between the primary and the secondary currents.
γ	Time angle of one-half switching cycle, measured by ω_o.
θ	Time angle for the circuit mode $sc(1,0)$ to appear, measured by ω_o'.
ω_o	The load-resonant angular frequency.
ω_o'	The transformer-resonant angular frequency.

B. Abbreviation

BCD	Border between operational domains of *CCM* and *DCM*1.
BDD	Border between operational domains of *DCM*1 and *DCM*2.
BDI	Border between operational domains of *DCM*2 and *IM*.
CCM	Continuous current operational mode.
DCMn	Discontinuous current operational mode n, where n is one or two.
FB-SRC	Full-bridge series-loaded resonant dc–dc converter.
IM	Idle operational mode.
LCT	Loosely coupled transformer.
SRC	Series-loaded resonant dc–dc converter.

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I. INTRODUCTION

ELECTRIC power can be transferred from one device to another either via conductive or inductive coupling. If one of the devices is mobile, this is the case when one charges an EV with ground charging facility, the transformer, in case of inductive coupling, has to be separable. In other words, the coupled inductors must be separately attached to each device respectively. High frequency is required to increase power

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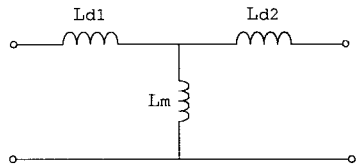


Fig. 1. Circuit model of LCT.

density of the inductors. The full-bridge SRC is chosen mainly based on the considerations as follows.

1) The primary side of the SRC can survive any accidental condition present at the secondary (secondary inductor + rectifier + load). This is desired by the aforementioned application because the secondary (vehicle side) will be the users' device which is probably due to a lack of professional maintenance.
2) The full-bridge structure is needed to obtain higher input dc source voltage as well as lower switches' blocking voltage, compared to its half-bridge counterpart. This choice is made on the considerations of the fact that the relatively high-leakage inductance of the LCT limits the minimum value of the resonant inductance. High-input voltage comforts the design of a high-power rate converter with large resonant inductance.

The analysis of the SRC was accomplished more than a decade ago [3]–[8]. All these publications were based on the assumption of unit coupling coefficient by assuming infinitive magnetizing impedance of the coupling transformer. Since then efforts have been made by many authors on analysis of the topologies including more than two resonant elements. Among these publications, [9] presented an orderly search procedure which leads to a number of three-element topologies. Reference [10] gives the analysis of the parallel resonant converter with the topology incorporating the "T" equivalent of the high-frequency transformer. The state plane analysis of an inductor parallel-loaded SRC of four-element topology is given in [11]. References [12] and [13] presented analysis of the LCL-type SRC with an inductive output filter which is essentially a four-element topology. References [14] and [15] include four-element LCL-type SRC topology with a capacitive output filter. In [14] and [15], analysis by using the fundamental approximation approach [16] is presented. All these analyzes of the SRC's dealt with the above resonance operation.

In this paper, the topology of the SRC containing a "T" equivalent of the transformer is investigated. Based on the application of battery charger, the load is presented as a voltage sink which, together with the diode bridge, is a current-driven rectifier. A "T" equivalent of the LCT as shown in Fig. 1 is used to build the circuit model of SRC. After per-unit normalization, the "T" equivalent is described by two parameters—the coupling coefficient k and the symmetry coefficient d. The novel contributions of the paper are summarized as follows.

1) Based on the parameters of the transformer, k and d, a novel operational plane of the operation above resonance is defined.
2) All the operational modes of the SRC which could appear in above resonance operation are identified.
3) Boundaries of each operational modes are defined, which in the two-dimensional (2-D) plane are functions of k.
4) Each of the operational modes is analyzed, and the experimental results are presented to verify the analysis.

After the nomenclature, the circuit models are created and solved in Section II, the analysis of the *CCM* is given in Section III, the analyzes of the discontinuous current operational mode 1 and 2 (*DCM*1 and *DCM*2) are given in Sections IV and V, respectively, the idle operational mode (*IM*) is presented in Section VI. The experimental results and the conclusions are given in Sections VII and VIII, respectively. All the results are given in per-unit normalized form.

II. Circuit Models and Solutions

In this section, the circuit model of an LCT is discussed. The parameters based on the "T" equivalent of the LCT are defined. This "T" equivalent is then incorporated into the SRC to form a circuit model of the LCT-based SRC. The time-domain solutions for the circuit modes which will appear in the operational modes are obtained. The expressions of the energy components during one-half switching cycle are derived.

A. Circuit Model and Parameters of the LCT

There are many special issues related to design, and modeling of an LCT taking into account the requirements of specific applications. At the level of converter analysis, assumptions will be made to simplify the problem, and only the most significant feature will be taken into account, which is the fact that the leakage inductances are comparable to the magnetizing inductance. The following assumptions have been made.

1) The magnetic circuits are linear.
2) The effect of distributed capacitances is negligible.
3) The losses in the transformer are negligible.
4) Turns ratio is unit.

Following the above assumptions leads to the generic "T" equivalent circuit for a two-winding transformer [10], [11], [17], which is shown in Fig. 1. Based on the "T" equivalent two parameters, the coupling coefficient k and the symmetry coefficient d are defined, respectively, as

$$k = L_m/\sqrt{L_1 L_2} \quad \text{and} \quad d = \sqrt{L_1/L_2} \tag{1}$$

where $L_1 = L_{d1} + L_m$ and $L_2 = L_{d2} + L_m$. From (1) the following expressions can be easily derived and they will be needed later:

$$k \le d \le 1/k \quad \text{and} \quad L_{s1}/L_1 = (1 - k^2) \tag{2}$$

where $L_{s1} = L_{d1} + L_{d2} || L_m$ is the primary inductance when the secondary is short circuited.

B. Circuit Model of the LCT-Based SRC

Fig. 2 shows the principle diagram of an LCT-based FB-SRC. The following assumptions are made to draw the equivalent circuit as given in Fig. 3 and to be used in the analysis.

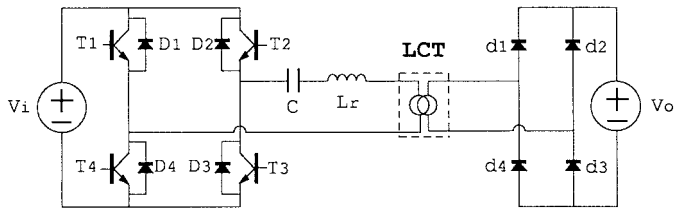


Fig. 2. An LCT-based FB-SRC for operating above resonance.

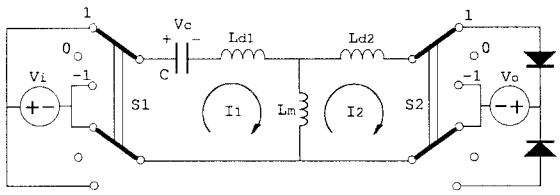


Fig. 3. Circuit model of the LCT-based SRC.

TABLE I
VALUES OF S_1 AND S_2

S_1	S_2	circuit condition
1		T_2,T_4 'on' or D_2,D_4 'on'
-1		T_1,T_3 'on' or D_1,D_3 'on'
0		T_1 ... T_4 'off' and D_1 ... D_4 'off'
	1	d_1,d_3 'on' ($I_2 > 0$)
	-1	d_2,d_4 'on' ($I_2 < 0$)
	0	d_1 ... d_4 'off' ($I_2 = 0$)

1) The input voltage source V_i and the load voltage sink V_o are ideal dc voltage sources.
2) The transistors, diodes, capacitance, and inductance are ideal components.
3) The LCT is defined by Fig. 1.
4) The resonant inductance L_r can be combined into either L_{d1} or L_{d2}, depending onto which side of the transformer it is put or whether it is put at all.

In Fig. 3 the inverter and the rectifier are simulated by three-position switches S_1 and S_2, respectively. Values are given to S_1 and S_2 in corresponding to their positions defined in Table I.

Each pair of the values of S_1 and S_2 represents a circuit mode which may appear in operation. Totally, there are nine circuit modes as listed in Table II(a). During one-half switching cycle, the conditions of $S_1 = 1$ and $S_1 = -1$ are mutually exclusive. Thus, only six out of the nine modes need to be considered, and they are listed in Table II(b). Notation $sc(S_1, S_2)$ is used to identify the circuit modes, where *sc* stands for subcircuit and S_1 and S_2 are the switching variables. The six circuit modes are drawn in Fig. 4.

C. Solutions for the Circuit Modes

When secondary side is conducting, the load-resonant frequency is

$$f_o = 1/2\pi\sqrt{L_{s1}C_r} \quad \text{and} \quad \omega_o = 2\pi f_o. \tag{3}$$

TABLE II
CIRCUIT MODES

S_1	1	1	1	0	0	0	-1	-1	-1
S_2	1	0	-1	1	0	-1	1	0	-1

(a)

1	1	1	0	0	0
1	0	-1	1	0	-1

(b)

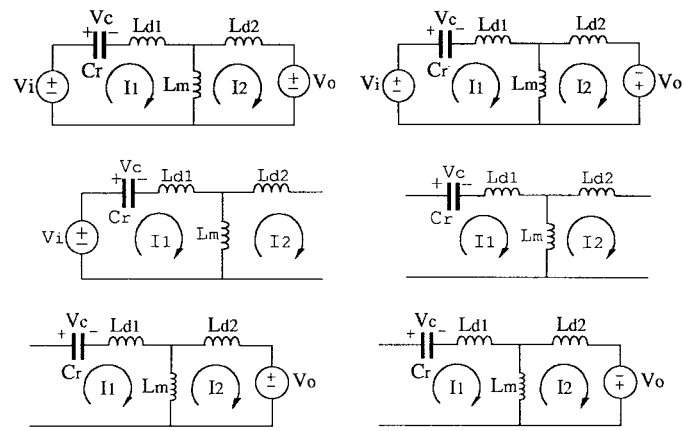


Fig. 4. Circuit modes.

When secondary side is not conducting, the transformer-resonant frequency is

$$f_o' = 1/2\pi\sqrt{L_1 C_r} \quad \text{and} \quad \omega_o' = 2\pi f_o'. \tag{4}$$

Applying (2) to (3) and (4) gives

$$\frac{f_o'}{f_o} = \frac{\omega_o'}{\omega_o} = \sqrt{1-k^2}. \tag{5}$$

The characteristic impedance is defined as

$$Z_o = \frac{1}{\sqrt{C_r L_{s1}}}. \tag{6}$$

Hereafter, all analytical terms will be given in per-unit normalized form without change in notation. The following quantities are chosen as the bases of the normalization:

voltage	V_i;
impedance	Z_o;
current	V_i/Z_o;
frequency	f_o;
power and energy	V_i^2/Z_o.

With the above preparation all the circuit modes in Fig. 4 can be solved. However, due to the fact that the primary is always conducting in above resonance operation, only $sc(1,1)$, $sc(1,-1)$, and $sc(1,0)$ will appear in the operation, and they will be solved in time-domain as follows.

1) Solution of sc(1, 1) and sc(1, −1): The circuit modes *sc*(1, 1) and *sc*(1, −1) are given in Fig. 4(a) and (b), respectively. The solution for them is

$$\begin{bmatrix} I_1(t') \\ I_2(t') \end{bmatrix} = \begin{bmatrix} \sin\omega_o t' & dk\sin\omega_o t' \\ dk\sin\omega_o t' & d^2(1-k^2)\omega_o t' + d^2k^2\sin\omega_o t' \end{bmatrix} \times \begin{bmatrix} 1 - V_c(t') \\ -q \end{bmatrix} + \begin{bmatrix} \cos\omega_o t' & 0 \\ -dk(1-\cos\omega_o t') & 1 \end{bmatrix} \times \begin{bmatrix} I_1(t_o) \\ I_2(t_o) \end{bmatrix} \tag{7}$$

$$V_c(t') = [1 - dkq - V_c(t_o)](1-\cos\omega_o t') + I_1(t_o)\sin\omega_o t') + V_c(t_o)$$

where $t' = t - t_o$ and $q = V_o/V_i$ which is the normalized load voltage or the conversion ratio. Obviously, to have the solution of *sc*(1, −1) we only need to change the sign of q in (7).

2) Solution of sc(1, 0): The scheme of the circuit mode *sc*(1, 0) is given in Fig. 4(c). The solution of *sc*(1, 0) is

$$\begin{aligned} I_1(t') &= \sqrt{1-k^2}[1 - V_c(t_o)]\sin\omega_o' t' + I_1(t_o)\cos\omega_o' t' \\ I_2(t') &= 0 \\ V_c(t') &= [1 - V_c(t_1)](1-\cos\omega_o' t') \\ &\quad + \frac{I_1(t_o)}{\sqrt{1-k^2}}\sin\omega_o' t' + V_c(t_o). \end{aligned} \tag{8}$$

D. Energy Conservation

1) Input Energy: The input energy during an interval (t_1, t_2) is

$$E_i = \int_{t1}^{t_2} I_1(t)\,dt. \tag{9}$$

The general form of the capacitor voltage is

$$V_c(t) = \omega_o \int_{to}^{t} I_1(t')\,dt' + V_c(t_o). \tag{10}$$

Comparing (9) and (10) gives

$$\omega_o E_i = V_c(t_2) - V_c(t_1). \tag{11}$$

2) Output Energy: The output energy during an interval (t_1, t_2) is

$$E_o = S_2 q \int_{t1}^{t_2} I_2(t)\,dt \tag{12}$$

where S_2 is the switching variable of the rectifier. Replacing $I_2(t)$ of (12) with the one of (7), (12) becomes

$$E_o = \frac{S_2 q'}{\omega_o}\left\{V_c(t_2) - V_c(t_1) + \left[\frac{I_2(t_1)}{dk} - I_1(t_1)\right]\omega_o(t_2 - t_1) - \frac{S_2}{K_q}2[\omega_o(t_2 - t_1)]^2\right\} \tag{13}$$

where

$$q' = dkq \quad \text{and} \quad K_q = \left(\frac{1}{k^2} - 1\right)q'. \tag{14}$$

Under steady-state condition for one-half switching cycle, we have

$$E_i = E_o. \tag{15}$$

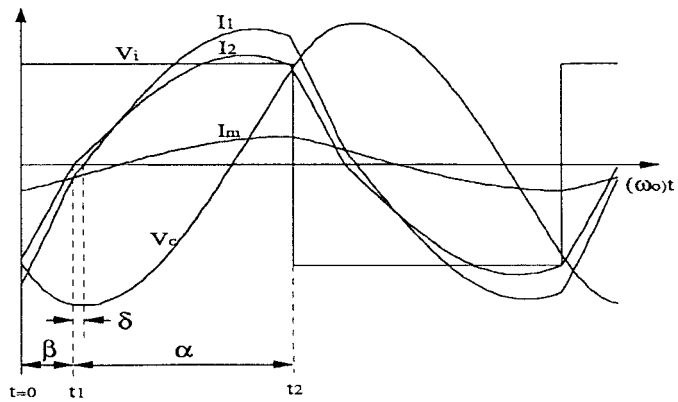


Fig. 5. Simulated waveforms of the CCM operational modes.

III. Continuous Current Operational Mode (CCM)

In this section, the continuous current operational mode is analyzed based on the waveform equations. In the procedure of the analysis, the normalized operational plane is determined with a novel parameter q' which is named transformed conversion ratio. Moreover, the increase of the reactive energy due to the fact of $k < 1$ is analytically evaluated.

A. Waveform Equations

The simulated *CCM* waveforms are shown in Fig. 5. The following notation is created to define an operational mode:

$$M\{sc(S_1, S_2), [sc(S_1, S_2)\ldots]\}$$

where the circuit modes appear in the order in which they appear in the operation of one-half switching cycle of the operational mode. Following this notation, *CCM* can be expressed as

$$M\{sc(1,-1), sc(1,1)\}. \tag{16}$$

By using (7) we can get the set of waveform equations of the Fig. 5 waveforms as follows:

$$\begin{aligned} I_1(t_1) &= [1 + q' - V_c(0)]\sin\beta + I_1(0)\cos\beta \\ I_2'(t_1) &= [1 + q' - V_c(0)]\sin\beta - I_1(0)(1-\cos\beta) \\ &\quad + K_q\beta + I_2'(0) \\ V_c(t_1) &= [1 + q' - V_c(0)](1-\cos\beta) + I_1(0)\sin\beta + V_c(0) \\ -I_1(0) &= [1 - q' - V_c(t_1)]\sin\alpha + I_1(t_1)\cos\alpha \\ -I_2'(0) &= [1 - q' - V_c(t_1)]\sin\alpha - I_1(t_1)(1-\cos\alpha) \\ &\quad - K_q\alpha + I_2'(t_1) \\ -V_c(0) &= [1 - q' - V_c(t_1)](1-\cos\alpha) + I_1(t_1)\sin\alpha \\ &\quad + V_c(t_1) \end{aligned} \tag{17}$$

where q' is defined by (14) which is named the transformed conversion ratio and $I_2'(t) = I_2(t)/dk$.

B. Energy Conservation and the Energy Factor

Using (11) to one-half switching cycle of *CCM* operation gives

$$\omega_o E_i = -2V_c(0). \tag{18}$$

By using (13) to the durations of β and α, respectively, and making the sum of the results, we obtain the output energy during one-half cycle of the *CCM* operation

$$\omega_o E_o = -2q' V_c(t_1). \tag{19}$$

The energy conservation [(15)] can be rewritten as

$$V_c(t_0) = q' V_c(t_1). \tag{20}$$

It is realized that the essential consequence of $k < 1$ is the increased reactive energy. This consequence is discussed hereafter. The energy flow within the converter during one-half switching cycle can be divided by concept into the following components:

- E_o energy into the load;
- E_i net energy into the inverter;
- E_e total amount of energy exchanged between V_i and the inverter;
- E_r the reactive energy.

Under steady-state condition

$$E_o = E_i \quad \text{and} \quad E_e = E_i + E_r. \tag{21}$$

It can be easily derived from (9) that

$$E_o = E_i = -2V_c(t_o)/\omega_o \quad \text{and} \quad E_e = 2V_{c\max}/\omega_o \tag{22}$$

where $V_{c\max}$ is the absolute value of the peak capacitor voltage. The energy factor [18] is defined as

$$EF = E_i/E_e. \tag{23}$$

For the converter with ideal transformer ($k = 1$), it is well known that $EF = q$ which is the conversion ratio. Considering the case of $k < 1$, from (23), (22), and (20) we have

$$EF = \left|\frac{V_c(0)}{V_{c\max}}\right| = q'\left|\frac{V_c(t_1)}{V_{c\max}}\right| = q \cdot (dk) \cdot \left|\frac{V_c(t_1)}{V_{c\max}}\right|. \tag{24}$$

The extreme right side of (24) shows that *EF* is a product of three terms. The first term q is based on the circulating energy caused by the conversion ratio. The second term dk is the effect of the coupling quality on the circulating energy via the transformed conversion ratio. The third term $|V_c(t_1)/V_{c\max}|$ represents the effect of the magnetizing energy.

C. Solution for the Time Angles

With the help of (20), the set of the waveform equations, (17), can be solved for the time angles as follows:

$$\sin^4\alpha + a_3\sin^3\alpha + a_2\sin^2\alpha + a_1\sin\alpha + a_o = 0 \tag{25}$$

where

$$a_3 = 2\frac{B}{A} \qquad a_2 = \frac{2(B^2 + A^2)}{A^2}$$
$$a_1 = \frac{2(AB - CD)}{-A^2} \qquad a_0 = \frac{(B^2 - D^2)}{-A^2}$$
$$A = -2q'$$
$$B = -\frac{q' K_q \gamma}{2}(1 + \cos\gamma) - (1 + q'^2)\sin\gamma$$
$$C = -\frac{q' K_q \gamma}{2}\sin\gamma - (1 + q'^2) - (1 - q'^2)\cos\gamma$$
$$D = -2q'\sin\gamma - \frac{q' K_q \gamma}{2}(1 + \cos\gamma)$$

and

$$\gamma = \alpha + \beta = \pi/f_s.$$

If we construct the operational plane with the transformed conversion ratio q' as the horizontal axis and the switching frequency f_s as the vertical axis, for any given point in *CCM* domain of the plane, the time angle α can be obtained from (25). Written in general form

$$\alpha = \alpha(f_s)@\{k, q'\}.$$

When α is obtained, all the waveform variables can be easily obtained from the waveform equations consequently.

In (25), if we let f_s approaching the resonant frequency

$$\lim_{f_s \to 1} \alpha(f_s)@\{k, q'\} = \pi - \sin^{-1}\sqrt{1 - q'^2}. \tag{26}$$

It is interesting to know from (26) that, at least on the horizontal line of $f_s = 1$, the boundary of the transformed conversion ratio is $q'_{\max} = 1$. Depending on the symmetry coefficient d [(2)], the boundary of the real conversion ratio q is $q_{\max} = 1/dk$ or $1 \le q_{\max} \le 1/k^2$. Thus, a novel normalized operational plane can be defined as

$$0 \le q' \le 1 \quad \text{and} \quad 1 \le f_s \le \infty. \tag{27}$$

1) Steady-State Analysis: Based on the obtained waveform variables, the steady-state operational terms can be easily calculated, and some of them are listed in Table III. The analytical results of the output power $P_o(f_s)$ when $k = 0.8$ and $k = 0.9$ are given in Fig. 10(a) and (b), respectively.

IV. Discontinuous Current Operation Mode 1 (*DCM*1)

In this section, the border between the operational domains of *CCM* and the discontinuous current mode 1 (*DCM*1) is analytically determined. In the domain of *DCM*1, the waveform equations are solved and steady-state analysis is obtained.

A. The Border Between the Operational Domains of CCM and DCM1 (BCD)

When f_s is raised to a certain level depends on the parameters q' and k, and the secondary side current is becoming discontinuous. The simulated waveforms of *DCM*1 are shown in Fig. 6. At the transition from *CCM* to *DCM*1, the operational mode is changing from

$$M\{sc(1,-1), sc(1,1)\}$$

to

$$M\{sc(1,-1), sc(1,0), sc(1,1)\}.$$

In *CCM* domain the secondary current during the interval $0 < t' < t_2 - t_1$, according to (7), is

$$I_2(t') = dk[1 - q' - Vc(t_1)]\sin\omega_o t' - dkI_1(t_1)(1 - \cos\omega_o t') - d^2(1 - k^2)\omega_o t'.$$

TABLE III
THE STEADY-STATE VARIABLES OF THE SRC

CCM	$-2V_c(0)\frac{f_s}{\pi}$	P_o
DCM1		
DCM2		
IM	0	
CCM	$-2V_c(0)\frac{f_s}{\pi q}$	I_o
DCM1		
DCM2		
IM	0	
CCM	$\lvert V_c(0)/V_{cmax}\rvert = qdk\lvert V_c(t_1)/V_{cmax}\rvert$	EF
DCM1		
DCM2	$\lvert V_c(0)/V_{cmax}\rvert$	
IM	0	
CCM	$V_{cmax} = \lvert V_c(t_1)\rvert + \lvert\Delta V\rvert$ where $\Delta V = [1-q'-V_c(t_1)](1-\cos\delta) + I_1(t_1)\sin\delta$ and $\delta = tg^{-1}(\frac{I_1(t_1)}{1-q'-V_c(t_1)})$	V_{cmax}
DCM1	$V_{cmax} = \lvert V_c(t_2)\rvert + \lvert\Delta V\rvert$ where $\Delta V = K_q(1-\cos\delta) + I_1(t_2)\sin\delta$ and $\delta = tg^{-1}(\frac{I_1(t_2)}{K_q})$	
DCM2	$V_{cmax} = \lvert V_c(t_1)\rvert + \lvert\Delta V\rvert$ where $\Delta V = K_q(1-\cos\delta) + I_1(t_1)\sin\delta$ and $\delta = tg^{-1}(\frac{I_1(t_1)}{K_q})$	
IM	$V_{cmax} = \lvert\Delta V\rvert$ where $\Delta V = (1-\cos\delta) + I_1(0)\sin\delta$ and $\delta = tg^{-1}(\frac{I_1(0)}{\sqrt{1-k^2}})$	

CCM	$(\frac{f_s}{2\pi})^{1/2}\{[1+q'-V_c(0)]^2(\beta-\frac{1}{2}\sin 2\beta)$ $+I_1^2(0)(\beta+\frac{1}{2}\sin 2\beta)+[1+q'-V_c(0)]I_1(0)(1-\cos 2\beta)$ $+[1-q'-V_c(t_1)]^2(\alpha-\frac{1}{2}\sin 2\alpha)+I_1^2(t_1)(\alpha+\frac{1}{2}\sin 2\alpha$ $+[1-q'-V_c(t_1)]I_1(t_1)(1-\cos 2\alpha)\}^{1/2}$	I_{1rms}
DCM1	$(\frac{f_s}{2\pi})^{1/2}\{[1+q'-V_c(0)]^2(\beta-\frac{1}{2}\sin 2\beta)$ $+I_1^2(0)(\beta+\frac{1}{2}\sin 2\beta)+[1+q'-V_c(0)]I_1(0)(1-\cos 2\beta)$ $+\sqrt{1-k^2}[1-V_c(t_1)]^2(\theta_1-\frac{1}{2}\sin 2\theta_1)+\frac{I_1^2(t_1)}{\sqrt{1-k^2}}$ $(\theta_1+\frac{1}{2}\sin 2\theta_1)+[1-V_c(t_1)]I_1(t_1)(1-\cos 2\theta_1)$ $+[1-q'-V_c(t_2)]^2(\alpha-\frac{1}{2}\sin 2\alpha)+I_1^2(t_2)(\alpha+\frac{1}{2}\sin 2\alpha)$ $+[1-q'-V_c(t_2)]I_1(t_2)(1-\cos 2\alpha)\}^{1/2}$	
DCM2	$(\frac{f_s}{2\pi})^{1/2}\{\sqrt{1-k^2}[1-V_c(0)]^2(\theta_1-\frac{1}{2}\sin 2\theta_1)$ $+\frac{I_1^2(0)}{\sqrt{1-k^2}}(\theta_1+\frac{1}{2}\sin 2\theta_1)+[1-V_c(0)]I_1(0)(1-\cos 2\theta_1)$ $+[1-q'-V_c(t_1)]^2(\alpha-\frac{1}{2}\sin 2\alpha)+I_1^2(t_1)(\alpha+\frac{1}{2}\sin 2\alpha)$ $+[1-q'-V_c(t_1)]I_1(t_1)(1-\cos 2\alpha)+\sqrt{1-k^2}[1-$ $V_c(t_2)]^2(\theta_2-\frac{1}{2}\sin 2\theta_2)+\frac{I_1^2(t_1)}{\sqrt{1-k^2}}(\theta_2+\frac{1}{2}\sin 2\theta_2)$ $+[1-V_c(t_2)]I_1(t_2)(1-\cos 2\theta_2)\}^{1/2}$	
IM	$(\frac{f_s}{2\pi})^{1/2}\{\sqrt{1-k^2}(\gamma-\frac{1}{2}\sin 2\gamma)$ $+\frac{I_1^2(0)}{\sqrt{1-k^2}}(\gamma+\frac{1}{2}\sin 2\gamma)+I_1(0)(1-\cos 2\gamma)\}^{1/2}$	

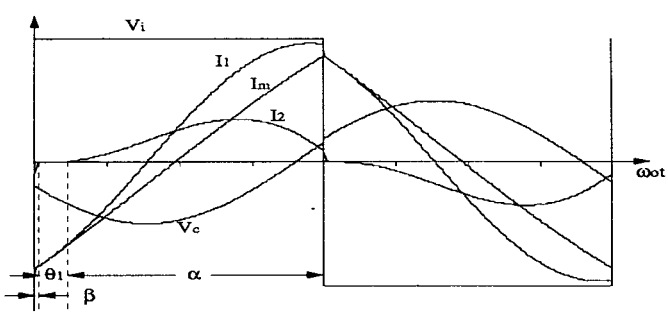


Fig. 6. Simulated waveforms of the DCM1 operational modes.

The transition of the circuit mode from $sc(1,-1)$ to $sc(1,0)$ requires

$$\lim_{t'\to 0} dI_2(t')/dt \le 0. \tag{28}$$

Equation (28) leads to the criterion of *DCM*1 which is

$$V_c(t_1) \ge 1 - q'/k^2. \tag{29}$$

By incorporating the condition of (29) into the set of *CCM* waveform equations (17), the following equations can be derived:

$$\cos\alpha + \frac{\gamma}{2}\sin\alpha = q'\frac{1-k^2/q'^2}{1-k^2} \tag{30}$$

and

$$\left(1+\frac{1}{k^2}\right)\cos\beta + \frac{\gamma}{2}\left(1-\frac{1}{k^2}\right)\sin\beta = \frac{1}{q'}\left(1+\frac{q'^2}{k^2}\right) \tag{31}$$

where

$$\gamma = \alpha + \beta.$$

The line BCD, described by $f_s(q')@(k)$ on the $f_s - q'$ plane, can be obtained by simultaneously solving (30) and (31). The result when $k = 0.863$ is given in Fig. 9. The existence of *DCM*1 is conditional, and it exists only when

$$q' > k^2. \tag{32}$$

Because the right side of (19) must be positive and $q' \ge 0$, there must be $V_c(t_1) < 0$. With this in mind, (29) needs

$q' > k^2$. As we can see in Fig. 9, the vertical line $q' = k^2$ is the asymptote of all the border lines.

B. Steady-State Analysis of DCM1

By using (7) and (8), the waveform equations of *DCM*1 can be obtained and from the waveform equations the following equations can be derived:

$$\sin\beta = \frac{K_q}{2}\left\{\sin(\alpha+\beta) - (\alpha+\beta)\cos(\alpha+\beta) - \left[(\alpha+\beta)\cos\theta_1 + \frac{\sin\theta_1}{\sqrt{1-k^2}}\right]\frac{1-\cos(\alpha+\beta)}{1+\cos\theta_1}\right\} \tag{33}$$

and

$$\cos\beta = \frac{K_q}{2}\left\{\frac{1+k^2}{1-k^2} + \cos(\alpha+\beta) + \frac{(\alpha+\beta)\sin(\alpha+\beta) - \frac{\sin\theta_1}{\sqrt{1-k^2}}[\sin(\alpha+\beta) + (\alpha+\beta)]}{1+\cos\theta_1}\right\} \tag{34}$$

where

$$\alpha+\beta = \gamma - \theta_1/\sqrt{1-k^2}.$$

In the operational domain of DCM1, the time angles α, β, and θ_1 can be obtained by solving the above equations, and then all the voltage and current variables can be calculated from the waveform equations. Based on the obtained waveform variables, the steady-state operational terms can be obtained, and some of them are listed in Table III. The analytical results of the output power $P_o(f_s)$ when $k = 0.8$ and $k = 0.9$ are given in Fig. 10(a) and (b), respectively.

V. Discontinuous Current Operation Mode 2 (*DCM*2)

In this section, the border between the operational domains of *DCM*1 and the discontinuous current mode 2 (*DCM*2) is analytically determined. In the operational domain of *DCM*2, the waveform equations are solved, and steady-state analysis is obtained.

A. The Border Between the Operational Domains of DCM1 and DCM2 (BDD)

The simulated waveforms of *DCM*2 are shown in Fig. 7. At the transition from *DCM*1 to *DCM*2, the operational mode is changing from

$$M\{sc(1,-1), sc(1,0), sc(1,1)\}$$

to

$$M\{sc(1,0), sc(1,1), sc(1,0)\}.$$

The condition for the transition from *DCM*1 to *DCM*2 is obviously $\beta = 0$. Putting this condition into (33) and (34)

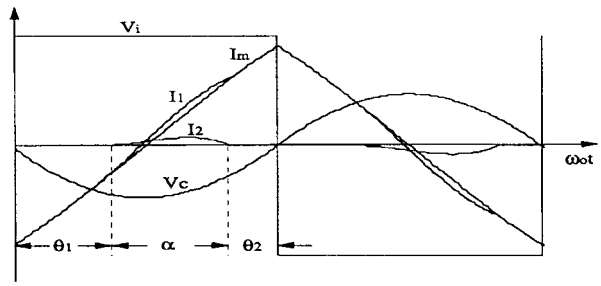


Fig. 7. Simulated waveforms of the DCM2 operational modes.

yields

$$\sin\alpha - \alpha\cos\alpha - \left(\alpha\cos\theta_1 + \frac{\sin\theta_1}{\sqrt{1-k^2}}\right)\frac{1-\cos\alpha}{1+\cos\theta_1} = 0 \tag{35}$$

and

$$\frac{1+k^2}{1-k^2} + \cos\alpha + \frac{\alpha\sin\alpha - \frac{\sin\theta_1}{\sqrt{1-k^2}}(\sin\alpha+\alpha)}{1+\cos\theta_1} = \frac{2}{K_q} \tag{36}$$

where

$$\theta_1 = \sqrt{1-k^2}(\gamma-\alpha).$$

The line *BDD*, described by $f_s(q')@(k)$ on the $f_s - q'$ plane, can be obtained by simultaneously solving (35) and (36). The result when $k = 0.863$ is given in the Fig. 9. The existence of DCM2 is conditional, and it exits only when $q' > k^2$.

B. Steady-State Analysis of DCM2

By using (7) and (8), the waveform equations of *DCM*2 can be obtained, and from the waveform equations the following equations can be derived:

$$2\sqrt{1-k^2}\sin\theta_2 - \frac{K_q}{\sqrt{1-k^2}}\sin(\theta_1+\theta_2) + \frac{K_q}{1-\cos\alpha} \times [(\sin\alpha-\alpha)\cos(\theta_1+\theta_2) + \sin\alpha - \alpha\cos\alpha] = 0 \tag{37}$$

and

$$2\cos\theta_2 - \frac{q'}{k^2}[1+\cos(\theta_1+\theta 2)] + K_q(1-\cos\alpha) + \frac{K_q(\sin\alpha-\alpha)}{1-\cos\alpha}\left[\sin\alpha - \frac{\sin(\theta_1+\theta 2)}{\sqrt{1-k^2}}\right] = 0 \tag{38}$$

where

$$\theta_1+\theta_2 = \sqrt{1-k^2}(\gamma-\alpha).$$

In the operational domain of *DCM*2, the time angles α, θ_1, and θ_2 can be obtained by solving the above equations, and then all the voltage and current variables can be calculated from the waveform equations. Based on the obtained waveform variables, the steady-state operational terms can be obtained, and some of them are listed in Table III. The analytical results of the output power $P_o(f_s)$ when $k = 0.8$ and $k = 0.9$ are given in Fig. 10(a) and (b), respectively.

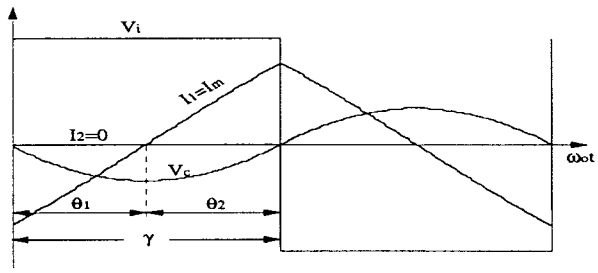


Fig. 8. Simulated waveforms of the IM operational modes.

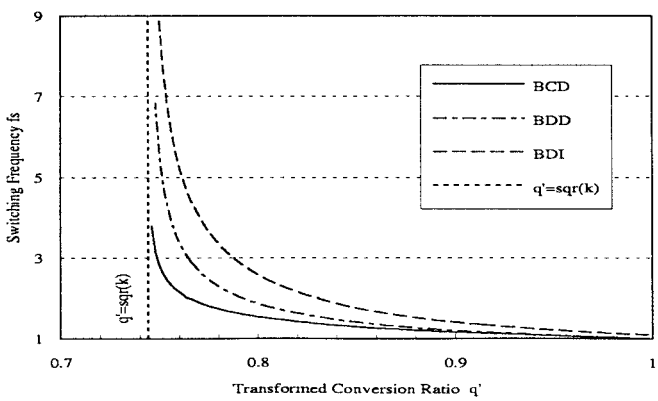


Fig. 9. The domains and the boundaries of all the operational modes.

VI. Idle Operation Mode (*IM*)

Operating in this mode, the output power of the converter is zero. In this section, the border between the operational domains of *DCM*2 and *IM* is analytically determined. The steady-state analysis is obtained.

A. The Border Between the Operational Domains of DCM2 and IM (BDI)

The simulated waveforms of *IM* are shown in Fig. 8(d). At the transition from DCM2 to IM, the operational mode is changing from

$$M\{sc(1,0), sc(1,1), sc(1,0)\}$$

to

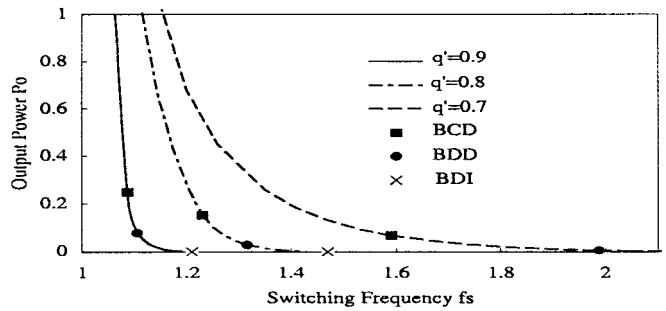
$$M\{sc(1,0)\}.$$

The condition for the transition from *DCM*1 to *DCM*2 is obviously $\alpha = 0$. By putting this condition into (37) and (38), the following equation can be derived:

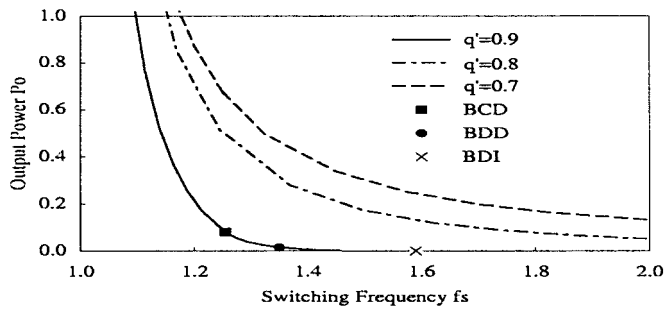
$$\cos\gamma = \frac{2k^4}{q'^2} - 1 \tag{39}$$

where $\gamma = \theta_1 + \theta_2$ and $f_s = \frac{\pi}{\gamma}\sqrt{1-k^2}$.

The line *BDI* can be obtained by solving (39). The result when $k = 0.863$ is given in Fig. 9. The existence of *IM* is conditional. It can be derived directly from (39) that *IM* exits only when $q' > k^2$. In Fig. 9, for given transformed conversion ratio q', the vertical distance between *BDI* and $f_s = 1$ is the switching frequency range needed for zero- to full-load output power control.



(a)



(b)

Fig. 10. Analytical results of P_o as function of f_s.

B. Steady-State Analysis of the IM

By using (8) the waveform equations of the *IM* can be obtained and from the waveform equations the following equations can be derived:

$$I_1(0) = \sqrt{1-k^2}\sin\gamma/(1+\cos\gamma) \tag{40}$$

and

$$V_c(0) = 0 \tag{41}$$

where $\gamma = (\pi/f_s)\sqrt{1-k^2}$.

In the operational domain of *IM*, the initial voltage and current can be calculated from (40) and (41). The steady-state operational terms can be obtained, and some of them are listed in Table III.

To give an example of the analytical results, power conversion characteristics $P_o(f_s)@\{k = 0.8, q' = 0.7, 0.8, 0.9\}$ and $P_o(f_s)@\{k = 0.9, q' = 0.7, 0.8, 0.9\}$ are plotted in Fig. 10(a) and (b), respectively.

VII. Experimental Verification

In this section, the design of a 600-W experimental converter is discussed. Tests were conducted with this converter and the results are compared to those of the analysis.

A. Design of the Experimental Converter

An FB-SRC with its primary and secondary coupled by an LCT is built for verifying the analytical results. The structure of the LCT is drawn in Fig. 11, where the core is considered primary side (ground device) with one side of

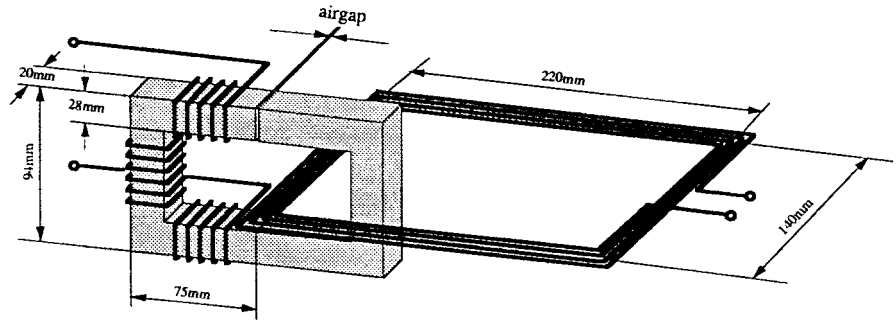


Fig. 11. Structure of the LCT for experiments.

TABLE IV
THE PARAMETERS OF THE LCT

air gap (mm)	0.0	0.25	0.5
L_m (μH)	1438	427.5	297.0
L_{d1}(μH)	26.5	17.6	16.0
L_{d2}(μH)	108.2	124.2	128.0
coupling coefficient k	0.956	0.863	0.814
symmetry coefficient d	0.974	0.898	0.858
turns ratio	16:4	16:4	16:4

TABLE V
THE PARAMETERS OF THE SRC

coupling coefficient k	0.956	0.863	0.814
resonant capacitance C_r (nF)	150	150	150
load-resonant frequency f_o (kHz)	36.4	0.38.5	40.0
trans.-resonant frequency f_o' (kHz)	10.7	19.5	23.2
characteristic impedance Z_o (Ω)	29.1	27.5	26.5
typical input voltage V_i (V)	100	100	100
typical output voltage V_o (V)	24	24	24
typical current per-unit base (A)	3.436	3.636	3.774
typical power per-unit base (W)	343.6	363.6	377.4

the rectangular flexible to let the secondary winding (vehicle device) get linked. The parameters at different airgaps are listed in Table IV where the parameters have been transformed to primary side. This loosely linked design is aimed to accommodate the vehicle positioning tolerance required by an automatic EV charging procedure which is briefly described as follows. The vehicle approaches the charging station and positions itself against the station with an alignment error within a specified tolerance. This tolerance is directly restricted by the size of the transformer core and the secondary winding. After the vehicle stands still, the secondary winding is moving out of the vehicle with the help of an actuator, pushing itself through the flexible leg of the core and getting coupled to the primary side.

Some of the converter design considerations are illustrated by the following case study.

The load is a set of lead–acid batteries with 24-V nominal voltage. The input supply is 100-V dc-regulated voltage source. The transformer turns ratio is 4:1 (16:4). The coupling coefficient k is measured 0.863, and the symmetry coefficient $d = 0.898$ when the airgap is 0.25 mm. The borders separating the operational domains can be calculated as shown in Fig. 9. The lowest battery voltage at charging is taken as $V_{o\min} = 24$ V (2.0 V per cell), and the highest is $V_{o\max} = 30.0$ V (2.5 V per cell). The maximum charging current is 30 A, and the minimum is 0 A. Thus, we can calculate the domain of the charging operation on the operational plane

$$q_{\min} = nV_{o\min}/V_i = 4 \times 24/100 = 0.98$$
$$q'_{\min} = dkq_{\min} = 0.898 \times 0.863 \times 0.98 = 0.744$$
$$q_{\max} = nV_{o\max}/V_i = 4 \times 30/100 = 1.2$$
$$q'_{\max} = dkq_{\max} = 0.898 \times 0.863 \times 1.2 = 0.93.$$

The departure of the above design $(0.98 < q < 1.2)$ from conventional analysis, in which we learned that the maximum value of the conversion ratio q shall not exceed 0.95 in one's design, becomes clear. Another new feature of the above design can be seen when we locate the operating points onto the operational plane. At the beginning of charging, we have $q' = 0.744$ and f_s close to one. At the end of charging, assuming charging current is zero, $q' = 0.93$ and the operating point must locate on the border line *BDI* where $f_s = 1.28$. It tells that the switching frequency range for whole charging procedure is $1 < f_s < 1.28$. This switching frequency range is considerably narrowed compared to the conventional analysis in which we learned $1 < f_s < \infty$.

The configuration of the converter is the same as shown in Fig. 2, and the parameters are listed in Table V.

91% dc-to-dc power efficiency was reached at these different transformer coupling coefficients.

B. The Operational Modes

All the operational modes have been observed, and the waveforms when $k = 0.863$ have been recorded and shown in Fig. 12. The location of the observed points on the operational plane is indicated in Fig. 13.

C. Verification of the Steady-State Analysis

1) Output Power: The analytical and the experimental results are plotted in Fig. 14. Because of the damping effect the experimental result is diverging from the analytical result

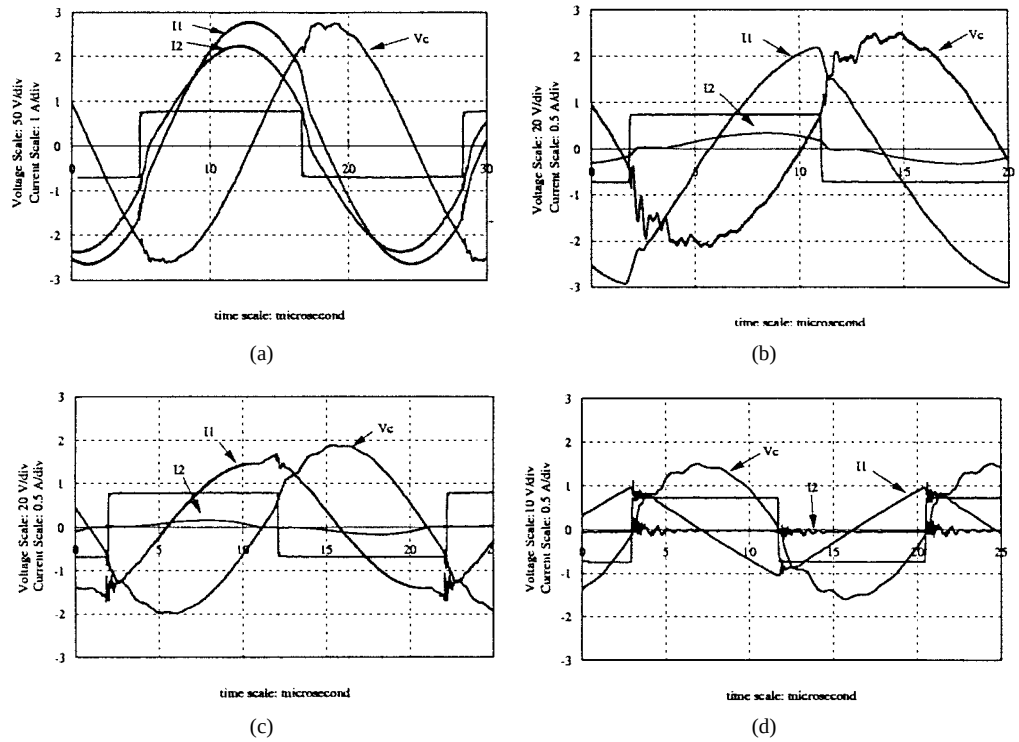


Fig. 12. Observed waveforms of all the operational modes when $k = 0.863$.

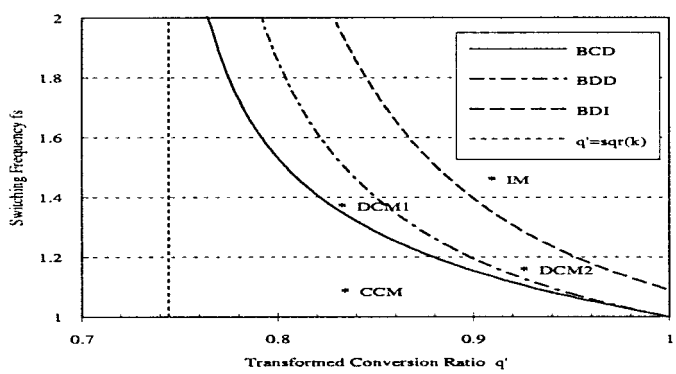


Fig. 13. Location of the observing points.

while the switching frequency is moving toward the resonant frequency. The results of numerical simulation are used to show the damping effect. In numerical simulation a lump resistance R is set in series with the resonant loop to simulate all the damping effects existing in the real circuit. The resonant quality is defined as $Q = R/Z_o$. The simulated results match analytical results perfectly when $Q = \infty$ and are gradually matching experimental result while Q decreasing.

Actually, the fundamental approximation method [14]–[16] is much simpler than the time domain method used here above. With the fundamental approximation method, the damping effect can be easily taken into account in analysis. However, the analytical results obtained in the analysis with the time domain method are beyond the reach of the fundamental approximation method. These results include the definition of the transformed conversion ratio q', determination of the boundaries of q' and q, construction of the new operational plane, the condition for the existing of IM, the clear depiction of the switching frequency range corresponding to output power regulation from full load to zero load, etc.

2) Peak Capacitor Voltage: The analytical, experimental, and simulating results of the peak capacitor voltage are put together in Fig. 15. The comments made for power conversion characteristics can also be applied to the results of $V_{c\max}$.

3) The Power Efficiency: Power efficiency is one of the final performances of a converter. It is interesting to see how the power efficiency is affected by the deterioration in coupling quality. The experiment is conducted in two scenarios. In the first scenario, the conventional operational plane is used, and the efficiency is measured at different coupling coefficients ($k = 0.956,\ 0.863$, and 0.814, respectively) while the real conversion ratio q is kept constant ($q = 0.89$ and $q' = 0.84,\ 0.69,\ 0.62$ corresponding to the different k). The results of these measurements are given in Fig. 16(a). In the second scenario, the redefined operational plane is used and the efficiency is measured at different coupling coefficients ($k = 0.956,\ 0.863$ and 0.814, respectively) while the transformed conversion ratio q' is kept constant ($q' = 0.84$ and $q = 0.89, 1.09, 1.20$ corresponding to the different k). The results of the measurements are given in Fig. 16(b). The input voltage and the circuit for the two sets of tests are exactly the same except the output dc voltage is manipulated by paralleling a variable resistive load to the battery at charging.

The use of a differently defined operational plane leads to a different design and consequently leads to significantly different power efficiency. The above tests are aimed to show that with different k, the conversion ratio q should and can be changed accordingly in the design to compensate the effect of k. The easiest way of doing this is to replace q with q'.

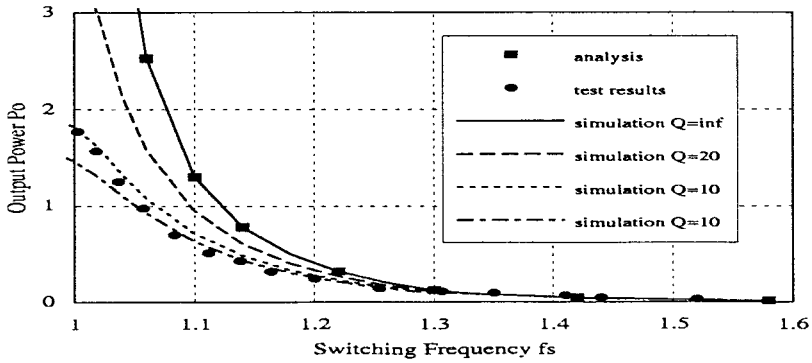


Fig. 14. Output power P_o versus switching frequency f_s.

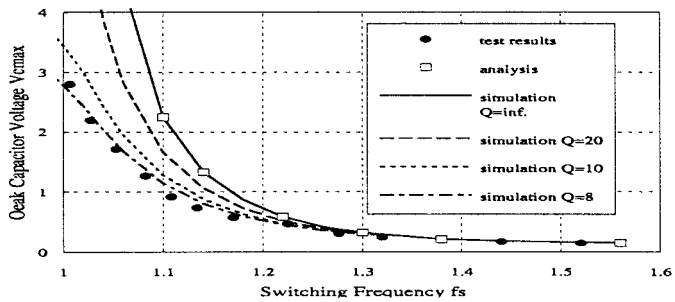


Fig. 15. Peak capacitor voltage $V_{c\max}$ versus switching frequency f_s.

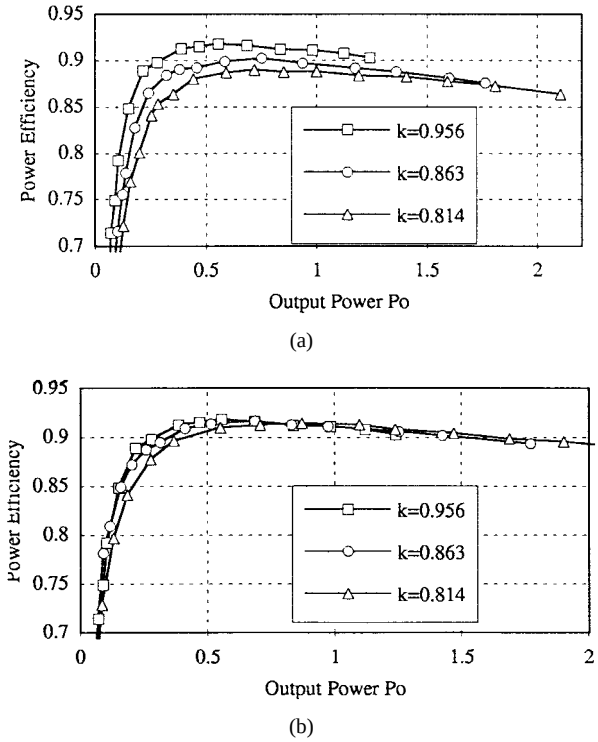


Fig. 16. Power efficiency measurements.

The mechanism behind this compensation can be explained with the energy factor *EF* defined by (23). The coupling quality affects the power efficiency mainly by decreasing the energy factor. According to (24), *EF* can be expressed as the product of two terms. The first term is $q' = dkq$ in which the effect of coupling quality on the conversion ratio is included. The second term, $V_c(t_1)/V_{c\max}$, represents the effect of magnetizing energy which depends on the coupling quality too. By keeping q' constant or, in another word, by changing q accordingly, the effect of the coupling quality on the conversion ratio is compensated. At high-power operation, this effect is dominant, hence, the the compensation is very effective as one can see in Fig. 16(b).

VIII. Conclusion

An LCT of the Fig. 1 circuit model can be characterized with two parameters k and d, which are, respectively, the coupling coefficient and symmetry coefficient. The steady state above resonance operation of an SRC in conjunction with such an LCT can be concluded as follows.

1) By defining the transformed conversion ratio q', a new operational plane can be defined which is $1 < f_s < \infty$ and $0 < q' < 1$.
2) There are four operational modes defined as *CCM*, *DCM*1, *DCM*2, and *IM*. The domains of the *DCM*1, *DCM*2, and *IM* locate to the right of the vertical line $q' = k^2$ in the operational plane.
3) Due to the existence of the *IM*, the frequency modulation characteristics are heavily dependent on k and q'. Given $q' > k^2$, the switching frequency range corresponding to full-load regulation is the vertical distance between $f_s = 1$ and *BDI*.
4) The circulating energy and the magnetizing energy are described by the energy factor *EF* which is defined by (23) in general and (24) for the *CCM* operation. By comparing (24) with $EF = q$, which is the result under the assumption of $k = 1$, it can be seen that the *EF* is affected by the coupling quality both through the conversion ratio (or circulating energy) and through the magnetizing energy. That means to compensate the effect of k, q should be changed accordingly in the design. The analysis shows the boundary of q does depend on k and can be set according to different k. The use of q' as a characteristic parameter to replace q opens the easiest way to achieve the compensation. The experimental results shows that the effect of k, regarding the energy factor and the power efficiency, can be significantly compensated, especially in high-power operation.

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